

ZIMNICEA, Romania

WMO: 154980

Lat: 43.6614N

Lon: 25.3536E

Elev: 35

StdP: 100.91

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-8.8	-13.1	1.2	-11.6	-10.2	1.6	-8.5	12.1	0.7	9.5	3.2	1.7	250	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	34.9	22.0	33.4	21.2	31.9	20.8	23.7	31.3	22.9	30.2	22.1	29.0	1.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.5	16.2	26.6	20.7	15.4	25.6	20.0	14.7	24.5	70.8	31.2	67.7	30.0	65.0	28.7	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.1	5.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	-1.1	3.5	7.9	12.3	17.9	22.3	23.9	24.0	19.6	12.5	6.4	2.2
	DBStd	9.35	5.00	4.80	4.22	3.97	3.15	2.70	2.50	2.65	3.54	3.95	4.73	3.88
	HDD10.0	1034	345	186	91	22	0	0	0	0	0	22	127	241
	HDD18.3	2643	603	415	324	182	46	3	1	1	28	184	357	499
	CDD10.0	2009	0	4	25	92	244	369	432	434	289	98	20	1
	CDD18.3	576	0	0	0	3	31	122	174	177	67	2	0	0
	CDH23.3	6234	0	0	6	86	368	1195	1831	1989	700	57	3	0
	CDH26.7	2663	0	0	0	14	90	470	831	987	262	9	0	0
Wind	WSAvg	2.1	2.1	2.2	2.3	2.4	2.1	2.0	1.9	1.8	2.0	1.9	1.7	2.4
Precipitation	PrecAvg	564	44	33	39	43	62	61	51	41	57	49	36	48
	PrecMax	907	119	99	79	85	136	170	167	130	166	124	100	157
	PrecMin	334	3	3	0	3	16	6	7	4	5	0	0	6
	PrecStd	137	25	21	20	21	29	37	40	32	41	33	24	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	19.9	24.2	28.5	32.3	35.8	36.9	36.8	34.1	27.6	22.3	15.8
		MCWB	8.3	14.2	15.2	17.7	21.0	23.8	24.5	21.8	19.8	16.7	15.3	11.3
	2%	DB	9.5	16.2	21.0	25.9	29.2	32.7	34.6	35.1	31.6	24.7	18.8	12.7
		MCWB	6.5	10.9	12.8	16.4	19.6	22.1	22.3	21.7	19.0	16.2	13.9	8.9
	5%	DB	7.1	13.3	18.4	23.0	27.6	31.2	33.0	33.5	29.5	22.2	16.0	10.0
		MCWB	5.2	9.7	11.7	15.1	18.7	21.7	21.7	20.7	18.5	15.2	12.6	7.3
	10%	DB	5.0	10.5	15.8	20.4	25.5	29.4	31.2	32.0	27.5	19.8	13.2	7.6
		MCWB	3.7	8.0	10.5	13.9	17.9	20.9	21.3	20.4	18.3	14.2	11.3	5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	14.7	15.4	18.7	22.2	24.6	25.0	24.6	22.2	18.2	15.7	11.5
		MCDB	12.1	19.9	23.2	26.4	30.4	32.5	33.4	32.7	30.0	23.9	20.5	14.5
	2%	WB	7.0	11.9	13.5	17.3	20.6	23.4	23.7	23.3	20.8	17.0	14.2	9.2
		MCDB	9.4	14.9	19.8	25.0	27.3	30.7	31.5	31.0	27.0	22.8	18.4	12.2
	5%	WB	5.4	9.9	12.0	15.9	19.6	22.4	22.9	22.4	19.7	16.0	13.1	7.7
		MCDB	6.7	13.3	17.5	22.5	25.6	29.3	30.0	29.7	26.0	21.7	15.7	9.9
	10%	WB	3.9	8.2	10.6	14.6	18.4	21.6	22.2	21.6	18.9	14.8	11.6	6.2
		MCDB	4.8	10.4	14.8	19.3	24.0	28.1	28.9	28.7	25.2	19.0	13.5	7.5
Mean Daily Temperature Range	5% DB	MDBR	6.9	9.3	10.7	12.1	11.8	12.3	12.7	14.0	12.8	11.6	7.5	6.6
		MCDBR	10.6	14.5	16.5	17.5	15.5	15.0	15.6	16.8	16.6	16.6	13.3	11.8
		MCWBR	7.9	9.5	9.5	8.9	6.8	5.5	5.0	5.1	5.8	8.5	7.7	8.3
	5% WB	MCDBR	9.5	13.5	16.0	16.4	13.8	13.7	13.1	13.9	13.9	14.8	11.3	11.5
		MCWBR	7.2	9.1	9.1	8.3	6.3	5.5	4.8	4.9	5.4	7.8	6.5	8.1
Clear-Sky Solar Irradiance	taub		0.348	0.379	0.418	0.470	0.445	0.453	0.463	0.476	0.441	0.405	0.373	0.352
	taud		2.280	2.210	2.152	2.023	2.171	2.187	2.165	2.124	2.186	2.271	2.313	2.292
	Ebn at Noon		755	793	810	796	832	824	810	783	778	757	720	710
	Edh at Noon		87	111	133	164	145	144	145	147	128	103	85	79
All-Sky Solar Radiation	RadAvg		1.48	2.28	3.43	4.75	5.87	6.44	6.52	5.83	4.24	2.75	1.57	1.20
	RadStd		0.22	0.30	0.33	0.49	0.47	0.44	0.43	0.35	0.33	0.31	0.22	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	+0.46	N/A	N/A	N/A	N/A	+0.41	N/A	-142	+69	N/A

Nomenclature: See separate page